

LEVERAGING PREDICTIBILITY OF DEEP LEARNING TECHNIQUES ON FINANCIAL TIME SERIES FORECASTING

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INTRODUCTION

Time series forecasting is a data science technique that uses historical data to predict future values. It's used in many industries, including finance, marketing, sales, and retail, to predict things like sales numbers, stock prices, weather, and consumer demand. Its characteristics include trend, cyclicity, seasonality, stationarity.

For a time series $X=\{X_1,X_2,...,X_t\}$, where each X_t is an observation at time t , the forecast for the next time period $t+1$ can be expressed as:

$$\hat{X}_{t+1} = \frac{1}{t} \sum_{i=1}^t X_i$$

This equation indicates that the forecast $\hat{X}_{t+1}$ is simply the average of all past observations up to time t . This approach assumes that future values will follow the same distribution as past values and is particularly useful when the time series does not exhibit trends or seasonality.

A financial time series is a sequence of financial data points, like stock prices, interest rates, or exchange rates, recorded at regular time intervals. It shows how these values change over time, helping investors and analysts track and predict market trends.

Financial time series forecasting is the process of predicting future financial data, such as stock prices or interest rates, based on past data patterns

Research in this field started almost 40 years back and since then there have been multiple advancements in this field.

ABSTRACT

Financial time series forecasting has become crucial in computational intelligence, engaging both academia and industry due to its broad applications. While many Machine Learning (ML) models have been explored, numerous surveys have already covered ML-based forecasting in finance.

Recently, Deep Learning (DL) models have shown notable improvements over traditional ML methods. Despite growing interest in DL for financial forecasting, research focused on DL applications for finance—especially region-specific data—remains limited. This study addresses this gap by experimenting with DL models to optimize forecasting accuracy for the Bombay Stock Exchange. Beyond reviewing current approaches, we aim to identify, improve, and potentially introduce a novel model tailored to Indian stock data, while also highlighting setbacks and opportunities for future DL research in financial forecasting.

PROBLEM STATEMENT

Financial time series forecasting faces significant challenges affecting prediction accuracy, including data quality issues, non-stationarity from market changes or global events, and inherent market volatility. These factors demand advanced modeling techniques to capture complex, nonlinear relationships beyond what traditional models can achieve.

Despite growing interest in deep learning (DL) for financial forecasting, comprehensive reviews on its effectiveness remain scarce. This research aims to address this gap by systematically reviewing DL methodologies, evaluating their strengths, weaknesses, and abilities to tackle challenges like overfitting and model generalization. Additionally, we experiment with DL models on Bombay Stock Exchange data to identify an optimized model for this context, ultimately enhancing accuracy and supporting better decision-making in finance.

LITERATURE REVIEW

Recent studies in financial time series forecasting demonstrate the strength of advanced machine learning and deep learning (DL) models. Traditional ML methods like ARIMA, GARCH, and SVM have limitations in handling non-linear patterns and long-term dependencies, which DL models are better equipped to address. For instance, LSTM models are favored for their capacity to capture complex patterns, especially when combined with adaptive features like batch normalization for improved accuracy during sharp market shifts (Fang et al., 2022). The Dual-LSTM model shows enhanced error reduction in predicting volatile markets (Gajamannage et al., 2023), while hybrid models such as Bi-GRU + CNN improve multivariable predictions, showing lowest RMSE values for Asian stock exchanges (Salgotra et al., 2024).

Other approaches have incorporated feature selection techniques (Niu et al., 2020) and graph neural networks (Cheng et al., 2021) to improve model interpretability and capture inter-stock dependencies. Despite these advances, gaps persist in model generalization and handling diverse data, particularly in region-specific applications like the Bombay Stock Exchange. Our study aims to fill this gap by developing a hybrid DL model combining CNN 3D and LSTM, targeting improved accuracy for Indian stock data.

The literature highlights deep learning models like LSTM, CNN, and hybrids as superior for capturing financial time series' complexities compared to traditional methods. However, gaps in model adaptability and scalability, especially for region-specific data, persist.

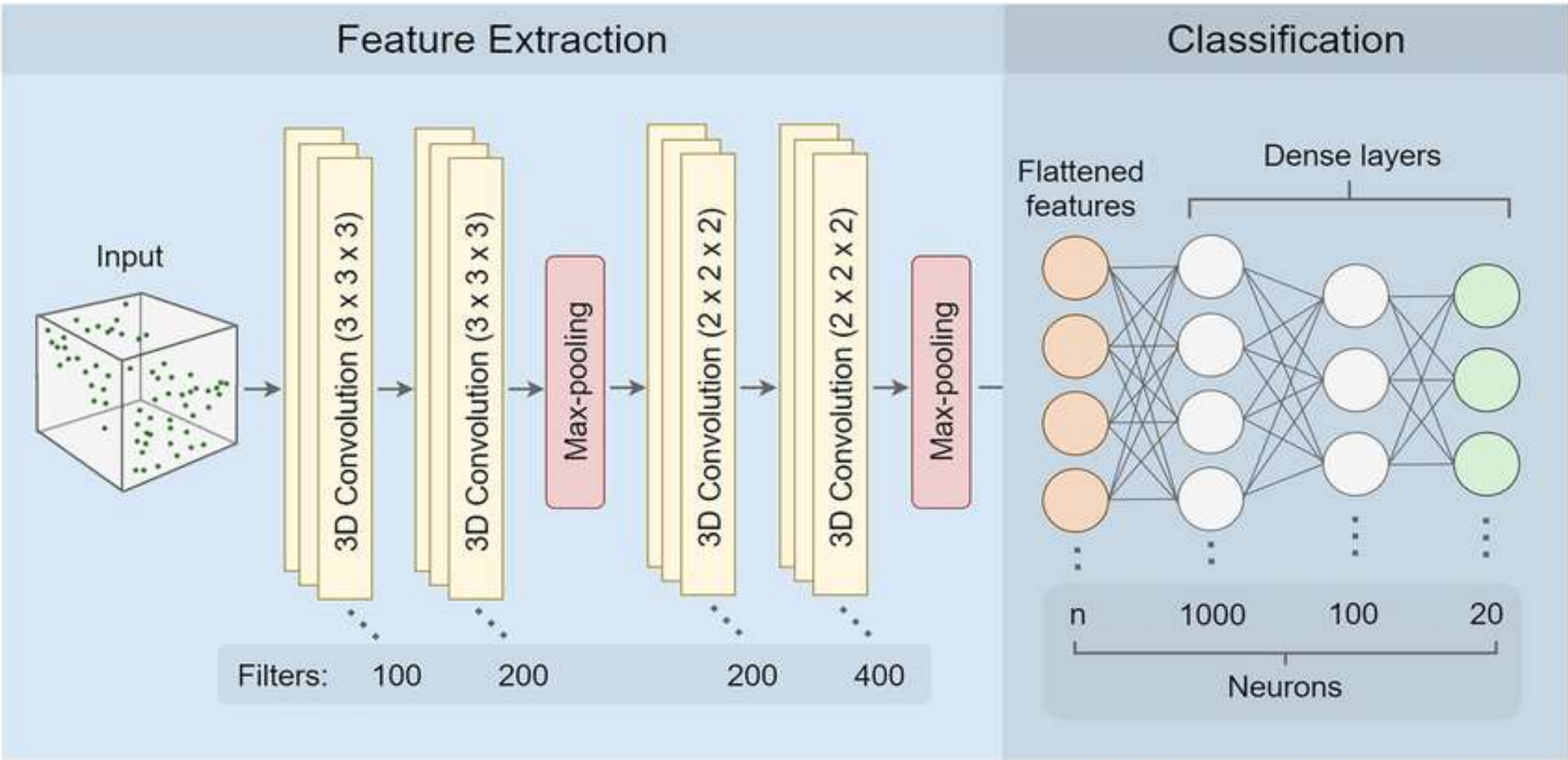
Detailed table with info of individual paper can be accessed through the following link: <https://tinyurl.com/research-table>

EXISTING ALGORITHM ANALYSIS

Model	MAE	MSE	RMSE	MAPE
ARIMA	15953.034	331024114.1	18194.068	18.604
SARIMA	1579.235	4.781	2186.705	1.87
CNN 1D	1595.166	3251346.784	1803.149	1.8555
CNN 2D	3808.437	17515250.51	4185.122	4.362
CNN 3D	1461.777	3466656.749	1861.896	1.7162
RNN	1488.511	2610015.624	1615.554	1.715
LSTM	1685.128	4919730.54	2218.046	1.978

The analysis of financial forecasting models using performance metrics (MAE, MSE, RMSE, and MAPE) shows varying effectiveness among algorithms. ARIMA exhibits the highest errors, indicating its limitations with complex financial data. SARIMA improves on ARIMA with lower errors but still lags behind deep learning models. Among these, CNN 1D shows relatively low error rates, making it a viable option, while CNN 3D outperforms all with the lowest errors, proving to be the best for capturing intricate financial patterns. CNN 2D has higher errors than CNN 1D and CNN 3D, suggesting it is less effective for time-series data. RNN shows competitive error rates, especially in MAE and RMSE, while LSTM performs well but is slightly less effective than CNN 3D.

PROPOSED MODEL



Our proposed model builds on CNN 3D as the core architecture, leveraging its capacity to capture spatial and temporal patterns in financial time series data. We aim to enhance forecasting accuracy by creating a hybrid model that integrates CNN 3D with LSTM, known for its strength in capturing long-term dependencies, or with other advanced DL models. This hybrid approach combines the strengths of spatial feature extraction and sequential pattern recognition, optimizing predictive performance specifically for financial forecasting on the Bombay Stock Exchange data.

CONCLUSION

In conclusion, integrating deep learning with traditional methods enhances financial forecasting by improving predictive accuracy and interpretability. Exploring hybrid models, attention mechanisms, and multi-task learning can significantly boost performance, while transfer learning and domain adaptation increase efficiency in specific financial domains.

Emphasizing explainable AI fosters trust and transparency, enabling stakeholders to understand model decision-making. Real-time forecasting through online learning and low-latency inference allows organizations to make informed decisions based on current data trends.

Addressing ethical considerations like fairness and bias is crucial for developing responsible AI-powered financial systems. This project sets the stage for future innovations in financial forecasting, merging advanced technology with ethical responsibility to meet the industry's evolving needs.

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